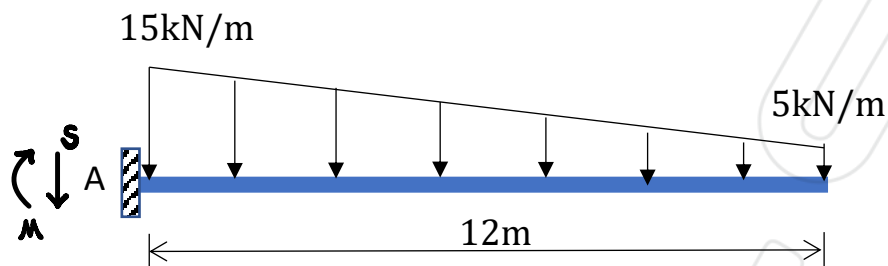


Structures I

Tutorial Sheet 4A: Shear and Bending Moment Diagrams

1. Determine the reactions on the beam shown below.

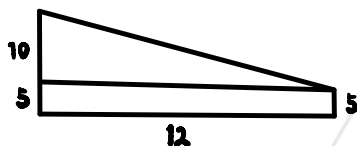


Ans: $R_{yA} = 120 \text{ kN}$; $M_A = 600 \text{ kN}\cdot\text{m}$

$$w = ax + b = 15 - \frac{5}{6}x \text{ (kN/m)}$$

$$\therefore R_{yA} = \int_0^{12} (15 - \frac{5}{6}x) dx = [15x - \frac{5}{12}x^2]_0^{12} = 120 \text{ (kN)}$$

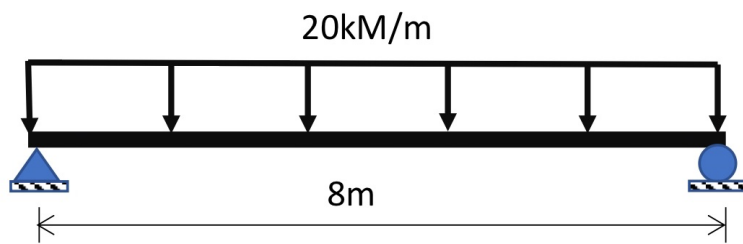
For , find centroid of x-axis



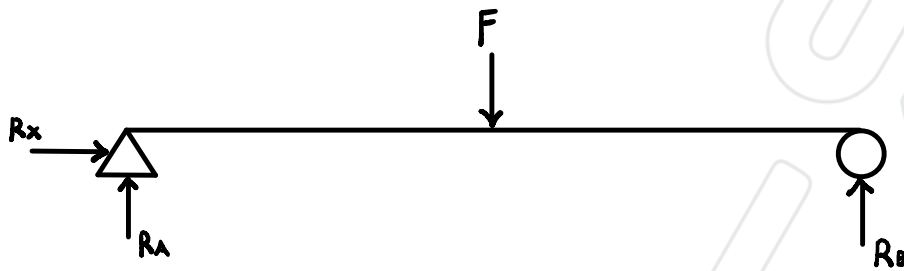
$$C_x = \frac{(\frac{1}{2} \times 12 \times (12 \times 5)) + (\frac{1}{2} \times 12 \times (\frac{1}{2} \times 12 \times 10))}{12 \times 5 + \frac{1}{2} \times 12 \times 10} = \frac{600}{120} = 5 \text{ (m)}$$

$$\therefore M_A = R_{yA} \cdot C_x = 600 \text{ (kN}\cdot\text{m)}$$

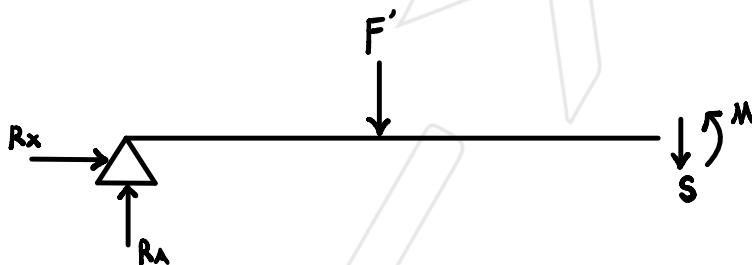
2. Draw the Shear and Bending Moment Diagrams for the beams shown below.



Ans: $S' = 80 - 20x$; $M = 80x - 10x^2$



$$R_A = R_B = \frac{1}{2}F = \frac{1}{2}(20 \text{ kN/m} \times 8 \text{ m}) = 80 \text{ kN}$$

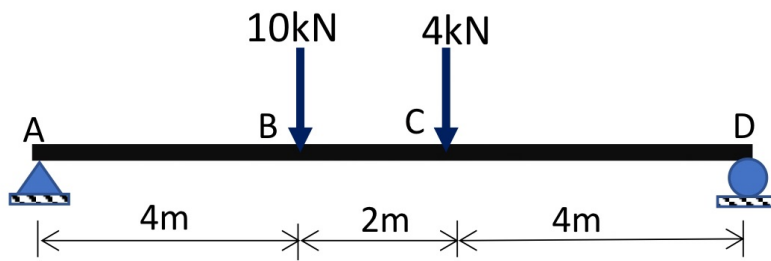


$$\therefore F' = 20x$$

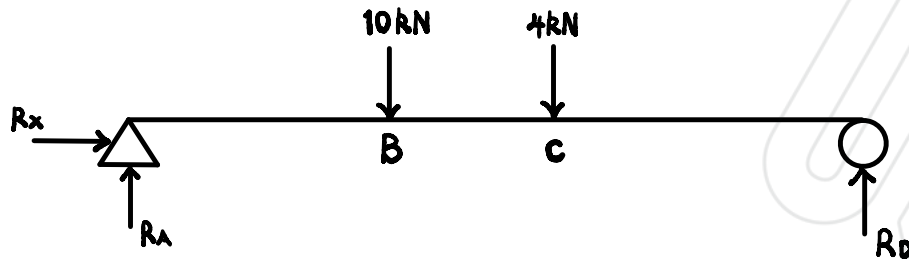
$$\therefore S(x) = 80 - 20x \text{ (kN)}$$

$$\therefore M(x) = 80x - 20x \left(\frac{x}{2} \right)$$

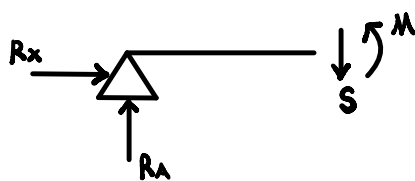
$$= 80x - 10x^2$$



Ans: $S = 7.6; -2.4; -6.4$ kN; $M = 7.6x; (40 - 2.4x); (-6.4x + 64)$ kN.m



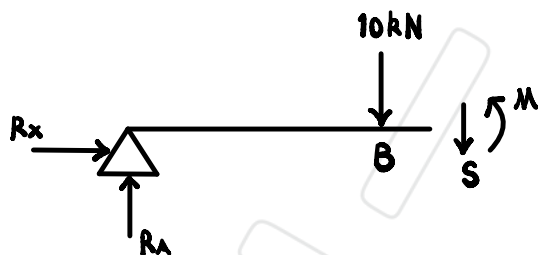
$$R_x = 0, \quad M(D) = 4 \times 4 + 10 \times 6 - R_A \times 10 = 0 \quad \therefore R_A = 7.6 \text{ (kN)}$$



$$0 \leq x \leq 4$$

$$S(x) = 7.6 \text{ (kN)}$$

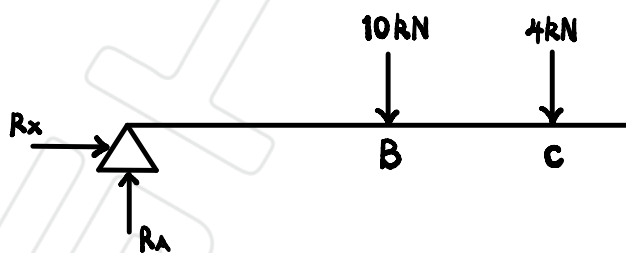
$$M(x) = 7.6x \text{ (kN} \cdot \text{m)}$$



$$4 \leq x \leq 6$$

$$S(x) = -2.4 \text{ (kN)}$$

$$M(x) = 7.6x - 10(x - 4) = -2.4x + 40 \text{ (kN} \cdot \text{m)}$$

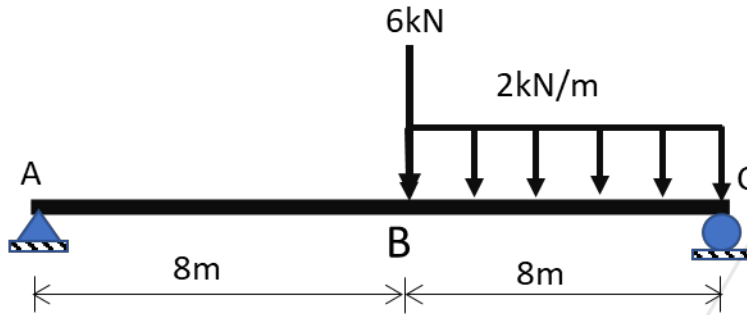


$$6 \leq x \leq 10$$

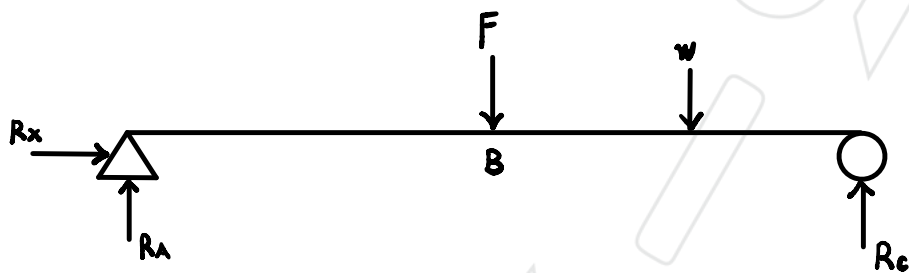
$$S(x) = -6.4 \text{ (kN)}$$

$$M(x) = -2.4x + 40 - 4(x - 6)$$

$$= -6.4x + 64 \text{ (kN} \cdot \text{m)}$$

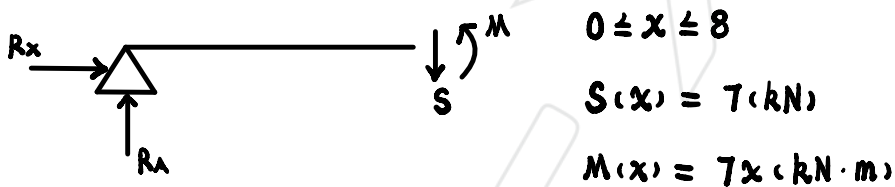


Ans: $S = 7; (17 - 2x) \text{ kN}; M = 7x; 7x - 6(x - 8) - (x - 8)^2 \text{ kN} \cdot \text{m}$



$R_x = 0 \text{ (kN)}$

$M(C) = 2 \times 8 \times 4 + 6 \times 8 - R_A \times 16 \quad \therefore R_A = 7 \text{ (kN)}$



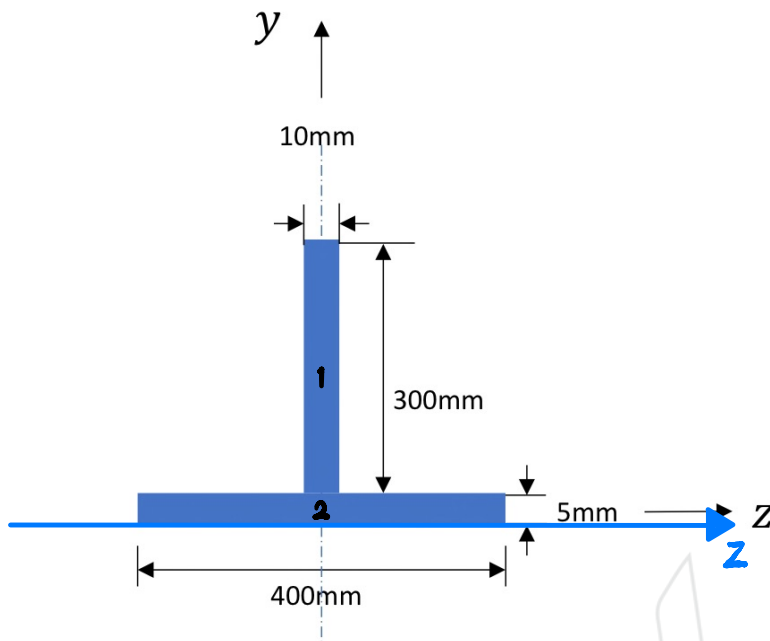
$8 \leq x \leq 16$

$S(x) = 7 + 6 + 2(x - 8) = 2x + 3 \text{ (kN)}$

$M(x) = 7x - 6(x - 8) - 2(x - 8) \frac{1}{2}(x - 8)$
 $= x^2 + 17x - 16 \text{ (kN} \cdot \text{m)}$

Tutorial Sheet 4B: Second Moments of Area

1. Evaluate the centroid and second moment of area I_x for the section beam shown below. Compare your solution to an equivalent section assumed as a thin-wall section.



Ans: $I_z = 50.6 \times 10^6 \text{ mm}^4$

$$y_G = \frac{(10 \times 300) \left(\frac{1}{2} \times 300 + \frac{1}{2} \times 5 \right) + (5 \times 400) \times 0}{10 \times 300 + 5 \times 400} = 91.5 \text{ (mm)}$$

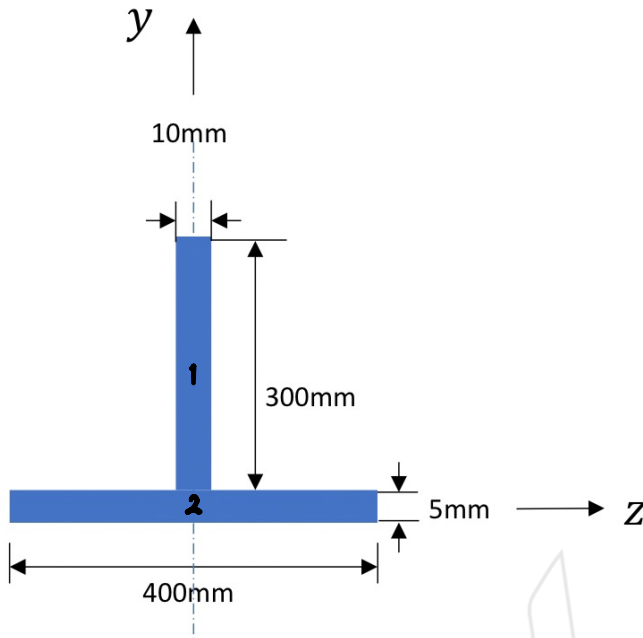
$$z_G = 0 \text{ (mm)}$$

$$y_G = \frac{(10 \times 300) \left(\frac{1}{2} \times 300 + 5 \right) + (5 \times 400) \times \left(\frac{1}{2} \times 5 \right)}{10 \times 300 + 5 \times 400} = 94 \text{ (mm)}$$

$$z_G = 0 \text{ (mm)}$$

Tutorial Sheet 4B: Second Moments of Area

1. Evaluate the centroid and second moment of area I_x for the section beam shown below. Compare your solution to an equivalent section assumed as a thin-wall section.



Ans: $I_z = 50.6 \times 10^6 \text{ mm}^4$

$$I_{y1} = \frac{300 \times 10^3}{12} = 25000 \text{ (mm}^4\text{)}$$

$$I_{y2} = \frac{5 \times 400^3}{12} = \frac{80000000}{3} \text{ (mm}^4\text{)}$$

$$\therefore I_y = 26.6916 \times 10^6 \text{ (mm}^4\text{)}$$

$$I_{z1} = \frac{300^3 \times 10}{12} + (300 \times 10) \left(300 \times \frac{1}{2} + 5 \times \left(\frac{1}{2} \right)^2 \right)^2 = 92268750 \text{ (mm}^4\text{)}$$

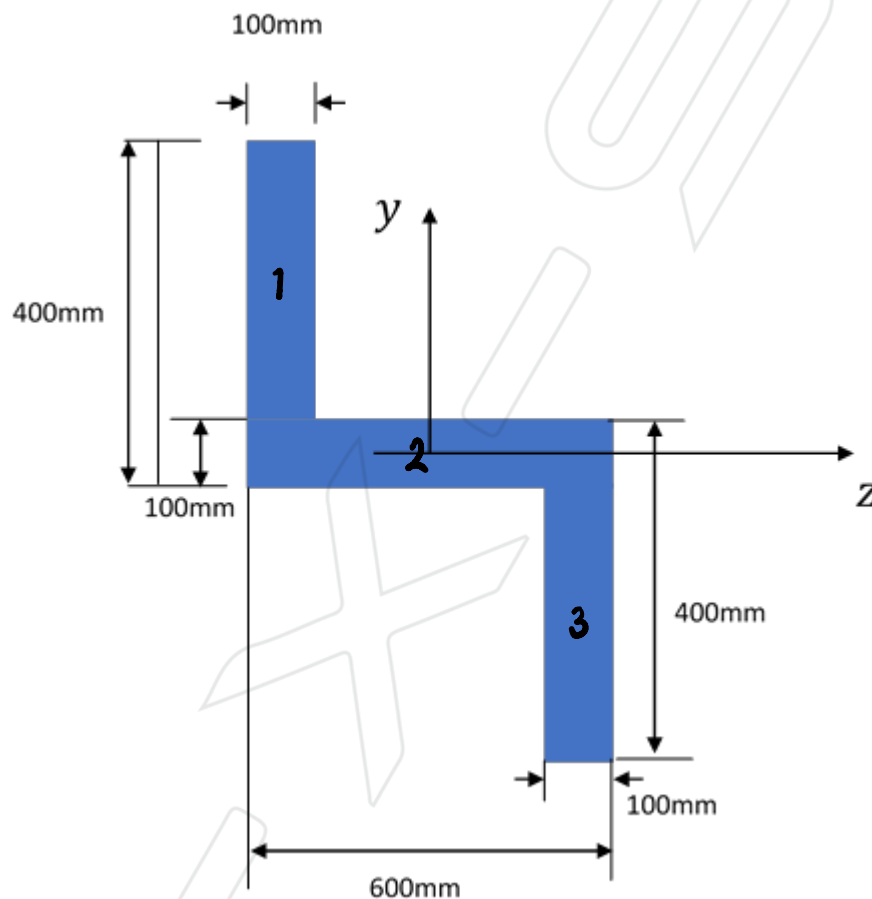
$$I_{z2} = \frac{400 \times 5^3}{12} + \frac{12500}{3} \text{ (mm}^4\text{)} + (400 \times 5) \times 5 \times \left(\frac{1}{2} - 94 \right)^2 = 16742666.67 \text{ (mm}^4\text{)}$$

$$\therefore I_z = 92.2729 \times 10^6 \text{ (mm}^4\text{)}, 50.41 \times 10^6 \text{ (mm}^4\text{)}$$

$$\therefore I_x = I_y + I_z = 118.9645 \times 10^6 \text{ (mm}^4\text{)}, 77.10 \times 10^6 \text{ (mm}^4\text{)}$$

2. Consider the beam section shown below.

- Evaluate the second moments of area I_z , I_y and I_{zy} .
- Determine the principal axis.
- Calculate the principal moments of area.



Ans: $I_z = 2.9 \times 10^9 \text{ mm}^4$; $I_y = 5.6 \times 10^9 \text{ mm}^4$; $I_{zy} = -3.0 \times 10^9 \text{ mm}^4$; $2\phi = 114.2 \text{ deg}$

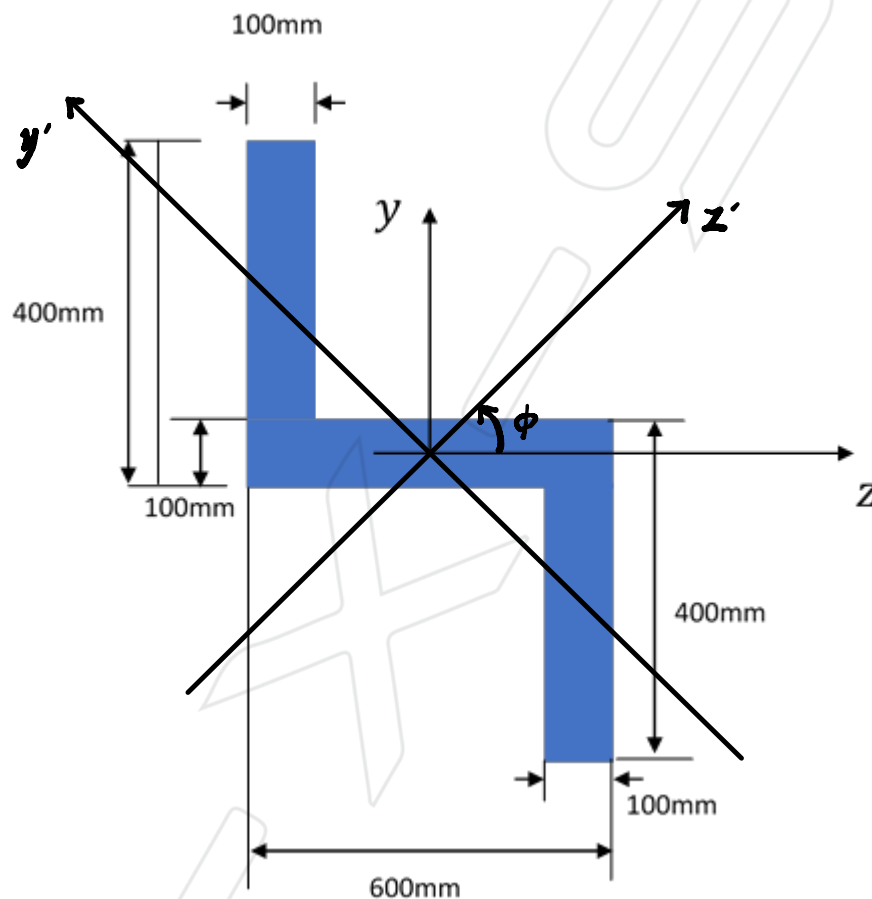
$$\begin{aligned} i) \quad I_z &= I_{z1} + I_{z2} + I_{z3} = I_{z2} + 2I_{z1} \\ &= \frac{600 \times 100^3}{12} + 2 \left[\frac{100 \times 300^3}{12} + (100 \times 300) \left(-\frac{1}{2} \times 300 - \frac{1}{2} \times 100 \right)^2 \right] = 2.9 \times 10^9 \text{ (mm}^4\text{)}, \end{aligned}$$

$$\begin{aligned} I_y &= I_{y1} + I_{y2} + I_{y3} = I_{y2} + 2I_{y1} \\ &= \frac{600^3 \times 100}{12} + 2 \left[\frac{100^3 \times 300}{12} + (100 \times 300) \left(\frac{1}{2} \times 600 - \frac{1}{2} \times 100 \right)^2 \right] = 5.6 \times 10^9 \text{ (mm}^4\text{)}, \end{aligned}$$

$$\begin{aligned} I_{zy} &= I_{zy1} + I_{zy2} + I_{zy3} = 0 + (100 \times 300) (250)(-200) + (100 \times 300) (-250)(200) \\ &= -3 \times 10^9 \text{ (mm}^4\text{)} \end{aligned}$$

2. Consider the beam section shown below.

- Evaluate the second moments of area I_z , I_y and I_{zy} .
- Determine the principal axis.
- Calculate the principal moments of area.



Ans: $I_z = 2.9 \times 10^9 \text{ mm}^4$; $I_y = 5.6 \times 10^9 \text{ mm}^4$; $I_{zy} = -3.0 \times 10^9 \text{ mm}^4$; $2\phi = 114.2 \text{ deg}$

ii) $I_{zy'} = 0$

iii) $\tan 2\phi = \frac{-2I_{zy}}{(I_z - I_y)} = \frac{-2 \times (-3 \times 10^9)}{2.9 \times 10^9 - 5.6 \times 10^9} = -\frac{20}{9}$

$\therefore 2\phi = \tan^{-1}\left(-\frac{20}{9}\right) = 114.23^\circ$